

Practice Quiz: Communication in Robotic Systems

Part A: Multiple Choice

1. Communication in a robotic Active Inference framework is best understood as: A) Only verbal speech between robots and humans B) The exchange of information between agents (robots, humans, or subsystems) that enables alignment of generative models and coordinated action C) The physical wiring connecting robot components D) Broadcasting raw sensor data to the internet

Answer: B -- Communication in Active Inference extends beyond data transmission to encompass the alignment of generative models between agents. When a robot signals its intended trajectory to a human coworker, it is helping the human update their model of the robot's future states, reducing mutual surprise.

1. In a multi-robot system, why is communication essential from an Active Inference perspective? A) Robots enjoy social interaction B) Each robot has a limited Markov blanket and cannot directly observe other robots' internal states; communication extends the effective sensory boundary to include information about other agents' beliefs and intentions C) Communication increases power consumption, which is always desirable D) It is not essential -- robots should always work independently

Answer: B -- Each robot's Markov blanket limits what it can directly observe. Communication acts as an extended sensory channel, allowing robots to share state estimates, predictions, and goals. This reduces collective free energy by enabling coordinated behavior.

1. A collaborative robot signals its intended motion to a human coworker through LED indicators and projected workspace boundaries. This is an example of: A) The robot decorating its workspace B) Legibility -- making the robot's predictions and intentions observable to the human, enabling the human to update their generative model of the robot's behavior C) The robot wasting energy on lights D) Communication that serves no functional purpose

Answer: B -- Legible robot behavior reduces the human's prediction error about what the robot will do next. Projected trajectories and LED state indicators externalize the robot's internal states, helping the human maintain an accurate model of the robot's plans.

1. In the ROS2 middleware, the DDS (Data Distribution Service) quality-of-service policies shape communication between nodes. In Active Inference terms, these policies define: A) The color scheme of the user interface B) The precision, reliability, and temporal characteristics of information flow across the Markov blankets of communicating subsystems C) The physical location of each node D) The robot's favorite communication protocol

Answer: B -- QoS policies (reliability, deadline, liveliness, history depth) determine how faithfully and quickly information flows between nodes. These correspond to the precision and bandwidth of the communication channel -- high reliability and low latency yield high-precision inter-subsystem information exchange.

1. When two mobile robots share their local occupancy grid maps over a communication channel, this is: A) Map merging -- combining partial generative models from different agents to form a more complete joint model with lower collective uncertainty B) A waste of bandwidth C) Only possible with identical robots D) An example of competitive behavior

Answer: A -- Each robot has built a partial map (local generative model) from its own observations. Sharing and merging these maps reduces both robots' uncertainty about the environment, lowering collective free energy -- a form of collaborative inference.

1. A human operator uses a joystick to teleoperate a robot in a hazardous environment. The communication channel here transmits: A) Only the robot's entertainment preferences B) Compressed action commands from the human's generative model to the robot's motor system, with video feedback returning sensory observations to the human -- forming a distributed perception-action loop across the communication channel C) Random signals D) Nothing useful

Answer: B -- Teleoperation creates a distributed Active Inference loop: the human's brain is the generative model, the communication channel bridges the human's decisions and the

robot's sensors, and the video feed closes the sensory loop. Latency and bandwidth limitations in the channel directly impact control performance.

1. Stigmergic communication -- where robots leave physical traces (markers, modified terrain) for other robots -- is analogous to: A) Littering B) Modifying the shared environment so that future sensory observations of other agents carry information, effectively using the world as a communication medium within the Active Inference framework C) Destroying the environment D) Communication that only works with humans

Answer: B -- Stigmergy (e.g., ants leaving pheromone trails) allows agents to communicate by modifying the external states that fall within other agents' Markov blankets. A robot marking a explored area with a physical tag changes what future robots will observe, indirectly transmitting information.

1. Communication bandwidth limitations affect Active Inference multi-robot systems by: A) Having no effect on system performance B) Constraining the precision of shared information, forcing robots to compress, prioritize, and selectively transmit the most informative aspects of their generative models C) Improving all aspects of coordination D) Only affecting audio communication

Answer: B -- Limited bandwidth means robots cannot share full state estimates. They must compress information (transmitting sufficient statistics rather than raw data), prioritize (sharing the most uncertain or decision-relevant variables first), and accept reduced precision on shared beliefs.

1. A robot asking a human "Should I pick up the red object or the blue one?" is an example of: A) The robot being indecisive without purpose B) Active communication to resolve ambiguity -- the robot's generative model cannot distinguish between two equally viable actions, so it seeks information from the human to reduce expected free energy C) The robot testing the human's color vision D) A programming error

Answer: B -- When the robot's policy evaluation produces a tie (equal expected free energy for multiple actions), querying the human provides additional preference information that disambiguates the decision. This is epistemic action through the communication channel.

1. Natural language communication between a human and a robot requires: A) The robot to have a generative model of the human's language, including how words map to world states, actions, and intentions -- essentially a shared generative model at the linguistic level B) Only a microphone and speaker C) The robot to understand every human language simultaneously D) No generative model of any kind

Answer: A -- Language understanding is generative model inversion: the robot must infer the speaker's intended meaning (hidden state) from the observed utterance (sensory data). This requires a generative model of how communicative intentions produce language, grounded in the shared physical and task context.

Part B: Short Answer and Design Prompts

1. Design a communication protocol for a team of three warehouse robots that must coordinate to avoid collisions while fulfilling orders. What information should each robot share, at what frequency, and how does this reduce collective free energy?
2. A robot arm and a human worker share a workbench. The robot must communicate its planned movements to the human. Compare three modalities: visual projection of planned trajectories, auditory beeps indicating movement onset, and haptic resistance when the human's hand enters the robot's planned path. Which provides the highest-precision information for the human's generative model?
3. Explain how communication latency in a teleoperation scenario increases free energy in the distributed human-robot system. What strategies (prediction displays, model-based compensation) can mitigate this?
4. How would you design a communication system for an underwater robot team where acoustic communication is available but severely bandwidth-limited? What aspects of the generative model should be prioritized for transmission?

5. A search-and-rescue robot encounters a survivor but cannot determine the best extraction path. Design an Active Inference communication strategy where the robot queries a remote human operator, incorporating the operator's response into its planning generative model.